KAIZEN MATHS

10-MINUTE PRACTICE

Algebraic substitution

Powers and products

Evaluate squares and products involving positive and negative values.

10 MINUTES / 5 QUESTIONS / AS-05

Name _____ Date _____

Show your working. Use brackets for negative values.

1 Given $a = 0, b = 0$

Evaluate $a^2 - 4ab + b^2$

2 Given $a = 1, b = 0$

Evaluate $a^2 - 4ab + b^2$

3 Given $a = 0, b = 2$

Evaluate $a^2 - 6ab + b^2$

4 Given $a = 1, b = 4$

Evaluate $a^2 - 2ab + b^2$

5 Given $a = 4, b = -1$

Evaluate $a^2 - 5ab + b^2$

Answer sheet

Powers and products | 5 questions

Keep this section separate from the question sheet. Full working follows on the next pages.

01 0

02 1

03 4

04 9

05 37

Worked solutions

Powers and products

01 Given $a = 0, b = 0$

$$a^2 - 4ab + b^2$$

Substitute $a = 0, b = 0$.

$$(0)^2 - 4(0)(0) + (0)^2$$

$$0$$

$$0$$

02 Given $a = 1, b = 0$

$$a^2 - 4ab + b^2$$

Substitute $a = 1, b = 0$.

$$(1)^2 - 4(1)(0) + (0)^2$$

$$1$$

$$1$$

03 Given $a = 0, b = 2$

$$a^2 - 6ab + b^2$$

Substitute $a = 0, b = 2$.

$$(0)^2 - 6(0)(2) + (2)^2$$

$$0 + 4$$

$$4$$

04 Given $a = 1, b = 4$

$$a^2 - 2ab + b^2$$

Substitute $a = 1, b = 4$.

$$(1)^2 - 2(1)(4) + (4)^2$$

$$1 - 8 + 16$$

$$9$$

Worked solutions

Powers and products

05 Given $a = 4$, $b = -1$

$$a^2 - 5ab + b^2$$

Substitute $a = 4$, $b = -1$.

$$(4)^2 - 5(4)(-1) + (-1)^2$$

$$16 + 20 + 1$$

$$37$$